Proficient in AutoCAD (Electrical) (80 Hours)				
S.No	**Session**	**Topics**	**Type**	**Duration (Hrs)**
1	SESSION 1	Introduction to engineering Drawings and AutoCAD Introduction to Engineering Drawings Various types of engineering drawings used in Electrical Industry (Pictorial diagram, Schematic diagram, One-line diagram, Block diagram, Wiring diagram) Introduction to AutoCAD Exploring GUI and Workspaces Co-ordinate systems **Display Control :** Zoom, Pan, Redraw, Regen, Clean Screen, Steering wheels **File Management:** New, Qnew, Open, Save, Save as, Close, Exit, Quit	Theory/Demo	2.0 Hrs
2	SESSION 2	Practice : Practice on Co - ordinate Systems	Practical	2.0 Hrs
3	SESSION 3	Drafting Basic Geometry Shapes in AutoCAD **Drawing settings:** Units, Limits Drafting settings Snap and Grid Polar tracking Object snap 3D Object snap Dynamic input (F1 to F12 Function keys) **Drawing tools:** Line, Circle, Arc, Ellipse, Donut, Polygon, Rectangle, Point, Multiple point	Theory/Demo	2.0 Hrs
4	SESSION 4	Practice : Practice on Drawing Tools	Practical	2.0 Hrs
5	SESSION 5	Drafting Basic Geometry Shapes in AutoCAD **Modify tools :** Erase, Undo, Redo, Explode, Move, Copy, Rotate, Mirror, Array, Align, Scale, Stretch, Lengthen, Trim, Extend, Break, Join, Chamfer, Fillet, Blend curves, Object selection **Object properties :** Properties window Color Linetype Linetype scale Line weight Match properties	Theory/Demo	2.0 Hrs
6	SESSION 6	Practice : Practice on Modify tools	Practical	2.0 Hrs
7	SESSION 7	Advanced Drafting and Modifying Tools in AutoCAD Drawing Tools Multiline Pline Spline Xline Ray Wipeout Revision cloud **Modify Tools :** Multiline style Multiline edit Polyline edit Spline edit, Offset PD Mode Grip editing PD Size Edit Array Change space Reverse Copy nested objects Delete duplicate objects Bring or send Objects **Views :** Orthographic, Isometric	Theory/Demo	2.0 Hrs
8	SESSION 8	Practice: Practice on Draw tools & Modify Tools	Practical	2.0 Hrs
		Layer Management		

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9	SESSION 9	Layer Management Adding / Removing Layers Layer Status New Property Filter New Group Filter Layer Status Manager Plot Control Object Selection Methods : Select, Qselect, Filter	Theory/Demo	2.0 Hrs
10	SESSION 10	Practice : Practice on Layers	Practical	2.0 Hrs
11	SESSION 11	Hatching Techniques **Hatching utilities:** Hatch Hatchedit Gradient Boundary Hatch Settings Fill and Fill mode **Isometric drawings:** Isometric snap Isometric drawings	Theory/Demo	2.0 Hrs
12	SESSION 12	Practice : Practice on Hatch & Isometric Drawing	Practical	2.0 Hrs
13	SESSION 13	Dimensioning, Tolerancing and Table Creation Techniques **Dimensions:** Linear, Aligned, Radius, Diameter, Angle, Arc length, Continuous, Baseline, Tolerance, Dimension Space, Dimension Break, Inspection, Jogged radius, Ordinate dimensions, Leader, Qleader, Oblique, Dimensions Text angle, Dimensions Text Alignment, Quick dimension Leader Multi leader Multi leader style Add leader Align Leader Collect Leader Remove Leader Annotation Tools Text Style Multiline Text Scale Text Spell Table Table style Table edit Dimension style Manager Dimension edit Dimension text edit Dimension Associative Re-associate Jog line Inspect Update	Theory/Demo	2.0 Hrs
14	SESSION 14	Practice: Practice on Dimension, Leader & Annotation	Practical	2.0 Hrs
15	SESSION 15	Application of Blocks and Attributes Create Block Base Write block Dynamic Block Insert **Attributes:** Attribute definition, Attribute edit, Edit single & multiple attribute, Attribute Display, Manage Attributes, Attdia, Attext, Eattext **Standards :** Drawing sheets standards and Title block standards Title blocks creation, Symbol creation, BOM creation, Data extraction and Link Data	Theory/Demo	2.0 Hrs
16	SESSION 16	Practice : Practice on Block & Attributes	Practical	2.0 Hrs

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17	SESSION 17	**External References:** Xref, Xbind, Circular Xref, Refedit Action Recorder **Express Tools:** Explode Attribute Arc Aligned Text Explode Text Case Torient Text Fit Text circle Modify text Super Hatch Object Linking and Embedding **OLE concepts:** Xref Xbind Circular Xref Refedit Copy Copybase Copylink Pasteclip Pastespecial Pasteblock Pasteoriginal	Theory/Demo	2.0 Hrs
18	SESSION 18	Practice :Practice on Xref & OLE	Practical	2.0 Hrs
19	SESSION 19	Layout, Plot and Publish Etransmit, Publish, Publish to web Layout, Page setup Viewports Viewport Lock and Unlock Viewport Layers Management Model space Paper space Plot styles and Plot (Print) in Model and Layout area	Theory/Demo	2.0 Hrs
20	SESSION 20	Practice : Practice on Layout & Plot	Practical	2.0 Hrs
21	SESSION 21	AUTOCAD ELECTRICAL AND PROJECT MANAGER Introduction AutoCAD Electrical GUI **Project:** Introduction to Project Manager, Working with Projects **Drawing:** Adding a Drawing, Creating a new Drawing, Draiwing Properties	Theory/ Demo	2 Hrs
22	SESSION 22	Practice: Practice on Drawing Creation	Practical	2 Hrs
23	SESSION 23	SYMBOL BUILDER Symbol Builder CIRCUIT BUILDER Circuit Builder Copy circuitry Save circuit to icon menu	Theory/ Demo	2 Hrs
24	SESSION 24	Practice: Practice on Symbol Builder , Circuit Builder	Practical	2 Hrs
25	SESSION 25	SCHEMATIC COMPONENT TOOLS Insert a component Connecting a component **Component Tools:** Insert a Components, Relocating Components, Inserting a Child Components, Aligning and Editing the components, Catalog information	Theory/ Demo	2 Hrs
26	SESSION 26	Practice: Practice on Inserting components & Editing	Practical	2 Hrs
27	SESSION 27	Component Attribute Tools WIRES AND WIRE NUMBERS TOOLS **Wires:** Wire layers, Wire types, Inert wire, Modify wire **Wire numbers:** Automatic wire numers, wire tagging, PLC I/O wire numbers	Theory/ Demo	2 Hrs
28	SESSION 28	Practice: Practice on Component Attribute & Editing Wires	Practical	2 Hrs
29	SESSION 29	**Wire numbers:** Automatic wire numers, wire tagging, PLC I/O wire numbers Wire number edit	Theory/ Demo	2 Hrs

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		Singnal Arrows: Source arrow, Destination arrow		
30	SESSION 30	Practice: Practice on Editing wires & Singnal Arrows	Practical	2 Hrs
31	SESSION 31	PROGRAMMABLE LOGIC CONTROLER(PLC) Ladder tools **PLC:** Generate PLC layout modules, PLC parametic selection, Module layout, Insert PLC modules modules, Edit PLC module, PLC database file	Theory/ Demo	2 Hrs
32	SESSION 32	Practice: Practice on PLC	Practical	2 Hrs
33	SESSION 33	CONNECTORS **Point to Point Wire Tools:** Introduction to connector diagrams, Inerting connectors, Editing & Modifyung connectors, Link components by dashed lines Grouping wires	Theory/ Demo	2 Hrs
34	SESSION 34	Practice: Practice on connectors	Practical	2 Hrs
35	SESSION 35	PANEL LAYOUT **Panel layout:** Foot Prints, Footprints from Schematic list, Footprints from icon menu, Din rails, Balloons, Wire Annotations, Create Assembly, Editing & Modifying Footprints, Creating Own Footprint	Theory/ Demo	2 Hrs
36	SESSION 36	Practice: Practice on Panel	Practical	2 Hrs
37	SESSION 37	**Terminals:** Placing a Terminal, Terminal Editor **Conversion tool:** Convert text, Convert block, Convert wires, Convert arrows, Special Explode	Theory/ Demo	2 Hrs
38	SESSION 38	Practice: Practice on Terminals & Conversion Tools	Practical	2 Hrs
39	SESSION 39	REPORTS AND CONVERSION TOOLS **Audit:** Missing Catalog, Electrical Audit, Signal Error/ List, Drawing Audit **Generate Reports:** Types of schematic reports, Generate a schematic report, Types of panel reports, Generate a panel report, Run automatic reports, Automatic report generation **Import/Export:** To Spreadsheet, From Spreadsheet	Theory/ Demo	2 Hrs
40	SESSION 40	Practice: Practice on Reports	Practical	2 Hrs